

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Electrical and Computer Engineering SEMESTER - 5 Winter 2025 ( Regular )**

**Course : Bachelor of Technology (Electrical and Computer Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 5**

**Subject Code & Name: BTECC501 - POWER SYSTEM-II**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The unit of reactive power is.  
a) Watt                      b) VA                      c) VAR                      d) Joule
  - 2 In reactance diagram, which elements are neglected?  
a) Resistance              b) Reactance              c) Capacitance              d) Inductance
  - 3 The main advantage of per unit system is...  
a) Reduction in calculations                      b) Uniformity of transformer parameters  
c) Elimination of losses                              d) Improved accuracy
  - 4 Load flow studies are mainly used to determine  
a) Fault current                                      b) Bus voltages and power flows  
c) Stability limit                                      d) Economic dispatch
  - 5 The primitive admittance matrix is always  
a) Symmetrical              b) Diagonal                      c) Sparse                      d) Singular
  - 6 The slack bus is required to  
a) Control reactive power                              b) Maintain system frequency  
c) Supply losses in the system                              d) Regulate voltage at all buses
  - 7 For transmission lines, the positive and negative sequence impedances are  
a) Always zero                      b) Equal                      c) Unequal                      d) Infinite
  - 8 Pre-fault voltage is usually assumed as  
a) Zero                      b) Rated value                      c) 0.8 pu                      d) 1.0 pu
  - 9 The most common fault in power systems is  
a) Line to line                                      b) Double line to ground  
c) Three phase fault                                      d) Single line to ground
  - 10 Steady-state stability deals with  
a) Large disturbances                                      b) Small and gradual disturbances  
c) Short-circuit faults                                      d) Rotor oscillations only

- 11 Power system stability refers to the ability of the system to  
 a) Operate at unity power factor    b) Maintain synchronism after disturbance  
 c) Deliver maximum power          d) Reduce losses
- 12 In a single-area LFC system, a sudden increase in load causes  
 a) Increase in frequency              b) Decrease in frequency  
 c) No change in frequency          d) Increase in voltage

Q2. Solve the following.

- A) What is the per unit system? Discuss its importance and advantages in power system calculations. 6
- B) Determine Ybus for the four bus system shown in figure the series impedance are determine Ybus by using direct inspection method. 6

| Line Bus | Impedance(PU) |
|----------|---------------|
| 1-2      | $0.25 + j1$   |
| 1-3      | $0.20 + j0.8$ |
| 1-4      | $0.30 + j1.2$ |
| 2-3      | $0.20 + j0.8$ |
| 3-4      | $0.15 + j0.6$ |

Q3. Solve the following.

- A) Define sequence impedance. Explain the positive, negative, and zero sequence networks of a transmission line 6
- B) Describe the single line-to-ground fault in a power system. Derive the boundary conditions and explain the interconnection of sequence networks with suitable diagrams. 6

Q4. Solve Any Two of the following.

- A) Describe the Gauss–Seidel method used for load flow analysis. Explain its advantages, disadvantages, and flowchart. 6
- B) Explain the Newton–Raphson method. State its advantages and disadvantages. Draw the flowchart of the method. 6
- C) Write a comparison between Gauss–Seidel, Newton–Raphson, and Fast Decoupled load flow methods. 6

Q5. Solve Any Two of the following.

- A) Enlist various factor affecting transient stability and explain methods to improve stage and trance in stability. 6
- B) Explain the sequence impedance of transmission lines and synchronous machines with neat diagrams. 6
- C) Draw and explain the sequence network of a star-connected load and its impact on fault analysis. 6

Q6. Solve Any Two of the following.

- A) Describe the role of a Load Dispatch Centre in a power system with a suitable block diagram. 6
- B) Describe the need for load frequency control and explain the LFC of a single-area system. 6
- C) What is an Automatic Voltage Controller? Explain its working and significance in maintaining generator voltage. 6

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